

Real Analysis I — Homework 1

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1. Let $\mathcal{F} = \{(a, b) \subset \mathbb{R} : a, b \in \mathbb{Q}, a < b\}$. Is $|\mathcal{F}| = \aleph_0$? If yes, find a one-to-one correspondence between $\mathcal{F}$ and $\mathbb{N}$. If not, prove your claim.
2. Suppose $E \subset \mathbb{R}$, and $|x - y| > 1$ for any $x, y \in E$. Prove that E is at most countable.
3. Suppose that A, B, C are sets, $A \subset B$, and $A \sim A \cup C$. Prove that $B \sim B \cup C$.
4. Suppose $A \subset B$. Show that $A' \subset B'$ and $\bar{A} \subset \bar{B}$.
5. Show that $\mathbb{R} \setminus \mathbb{Q}$ is a G_δ -set.
6. Suppose (X, d) is a metric space and $E \subset X$. Prove that the following three statements are equivalent:
 - (a) $x_0 \in E'$.
 - (b) There exists a sequence $\{x_k\} \subset E$, $x_k \neq x_0$ for all $k \in \mathbb{N}$, such that $x_k \rightarrow x_0$.
 - (c) For any open set U such that $x_0 \in U$ (U is called an open neighborhood of x_0), U contains infinitely many points of E .
7. Suppose $\{f_j\}$ is a sequence of functions where $f_j : \mathbb{R}^n \rightarrow \mathbb{R}$. Define $E_{j,k} = \{x \in \mathbb{R}^n : f_j(x) \geq \frac{1}{k}\}$ for $j, k \in \mathbb{N}$. Express the set $\{x \in \mathbb{R}^n : \limsup_{j \rightarrow \infty} f_j(x) > 0\}$ in terms of $E_{j,k}$'s.
8. Suppose $\{A_k\}$ and $\{B_k\}$ are two sequences of sets. Prove that
$$\limsup_{k \rightarrow \infty} (A_k \cup B_k) = \left(\limsup_{k \rightarrow \infty} A_k \right) \cup \left(\limsup_{k \rightarrow \infty} B_k \right)$$
9. Suppose G_1 and G_2 are open sets in $\mathbb{R}^n$, and $G_1 \cap G_2 = \emptyset$. Show that $G_1 \cap \bar{G}_2 = \emptyset$.
10. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is a monotonically non-decreasing function, and $E = \{x \in \mathbb{R} : f(x + \epsilon) - f(x - \epsilon) > 0, \forall \epsilon > 0\}$. Show that E is a closed subset of $\mathbb{R}$.